

1 Structured summary: A combined data-generating architecture
2 for biomarker-treatment interactions: channel super-additivity
3 and mixing-weight sensitivity in aggregated N-of-1 trials

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17 *This document is a structured, section-by-section summary of the key points argued in the*
18 *companion manuscript, “A combined data-generating architecture for biomarker-treatment in-*
19 *teractions: channel super-additivity and mixing-weight sensitivity in aggregated N-of-1 trials.”*
20 *It is provided as a supplementary reading aid and does not stand on its own; all notation,*
21 *derivations, simulation detail, and citations are in the main manuscript.*

22

23 **1 1. Introduction**

24 **The companion paper contrasts two data-generating process (DGP) architectures;**
25 **this paper treats the case where both operate simultaneously.**

- 26 • The mean-moderation architecture encodes the interaction in the mean; the covariance-
27 moderation architecture encodes it in the covariance.
- 28 • A biomarker could act through both pathways at once.

- The dual-channel architecture carries independent weights on the two channels and recovers A and B as its single-channel limits.

The paper makes two contributions: an analytical super-additivity result and a calibrated simulation of the combined grid.

- Analytical: the channel signals add, and convexity of the power function makes their power combination super-additive.
- Empirical: a 3×3 channel-weight grid crossed with design and carryover, with boundary cells validated against the companion paper's pure-architecture baselines.
- Guidance: when a combined model is warranted and how to use it as a sensitivity framework.

1.1 2.1 Definition

The dual-channel architecture applies both interaction mechanisms in sequence: a differential-correlation multivariate normal (MVN) draw, then an on-drug mean shift.

- The MVN covariance encodes BM-BR differential correlation with weight $c_{bm,B}$ (the covariance-moderation architecture channel).
- After the draw, BR is additively shifted on-drug by $c_{bm,A} \cdot z_B \cdot \sigma_{BR}$ (the mean-moderation architecture channel).
- The two weights are independent and are calibrated separately.

1.2 2.2 Boundary reductions and identifiability

The two channels are algebraically independent, so the boundary cases recover the pure architectures exactly.

- $c_{bm,B} = 0$ removes the covariance channel and recovers the pure mean-moderation architecture.
- $c_{bm,A} = 0$ removes the mean-shift channel and recovers the pure covariance-moderation architecture.
- The boundaries are enforced by construction, so they serve as internal validation gates.

1.3 2.3 Biological rationale

The two channels are mechanistically distinct and need not scale in proportion.

- The mean-shift channel (A) reflects pharmacodynamic scaling: the drug's effect is larger in high-biomarker participants.
- The covariance channel (B) reflects differential co-regulation that couples biomarker and response on-drug and washes out off-drug.
- Neither channel implies the other; a drug could activate one, both, or neither.

2 3. Channel Super-additivity: an Analytical Account

The biomarker-treatment signal delivered to the analysis model is additive across the two channels.

- The interaction coefficient depends on the on-drug covariance $\text{Cov}(B, BR)$.
- The covariance channel contributes $c_{bm,B} \sigma_B \sigma_{BR}$; the mean channel contributes $c_{bm,A} \sigma_B \sigma_{BR}$.
- The total on-drug signal covariance is $(c_{bm,A} + c_{bm,B}) \sigma_B \sigma_{BR}$; the channels add.

Additive signal becomes super-additive power because the power function is convex over the sub-ceiling regime.

- Power is a convex function of the noncentrality where power is below roughly one half.
- Convexity gives $\pi(\eta_A + \eta_B) - \pi(0) > [\pi(\eta_A) - \pi(0)] + [\pi(\eta_B) - \pi(0)]$.
- Hence interior cells exceed the probability-scale additive benchmark; the effect attenuates and can reverse near the ceiling.

2.1 4.1 Design

A 3×3 channel-weight grid crossed with three designs and three carryover half-lives.

- Channel weights: $c_{bm,A} \times c_{bm,B} \in \{0, 0.22, 0.45\}^2$.
- Designs: CO, Hybrid, OL+BDC; carryover $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks.
- E2 analysis specification; 1000 replicates per cell.

2.2 4.2 Boundary validation against the companion baselines

The boundary cells are validated numerically against the companion paper's pure-architecture power values.

- The combined driver is run under the same sample-size and calibration settings as the companion paper's Section 3.1 driver.
- The $c_{bm,B} = 0$ column must reproduce the pure mean-moderation architecture; the $c_{bm,A} = 0$ row must reproduce the pure covariance-moderation architecture.
- Agreement to within Monte Carlo error is the validation gate.

2.3 4.3 Empirical power over the channel-weight grid

Power is monotone in both channel weights, and interior cells are super-additive, as Section 3 predicts.

- Power increases strictly with each channel weight, holding the other fixed, across all three designs.
- Interior cells exceed the probability-scale additive benchmark $p(c_A, 0) + p(0, c_B) - p(0, 0)$.
- The design ranking (Hybrid > OL+BDC > CO) is preserved throughout the grid.

97 3 5. When a Combined Architecture is Appropriate

98 The dual-channel architecture is appropriate when both a pharmacodynamic-
99 scaling pathway and a co-regulatory coupling pathway are plausible for the same
100 biomarker.

- 101 • Use it when independent evidence supports both channels, not as a default.
- 102 • When channel weights are unknown, the α -mixing parameterization spans pure A to
103 pure B as a sensitivity analysis.
- 104 • It carries two free parameters where one suffices, so calibration demands more prior
105 information.

106 4 6. Discussion

107 The combined architecture unifies the companion paper's two architectures and
108 explains their descriptive combined results.

- 109 • A and B are single-channel limits of one model, not arbitrary alternatives.
- 110 • Super-additivity is a property of the power function, not an artifact of the calibration.
- 111 • The analysis model is held fixed, so calibration questions are deferred to the companion
112 calibration study [[@pmsimstats-paper10](#)].